Nutritional and Functional Characteristics of Whey Proteins in Food Products

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ABSTRACT

Whey proteins are well known for their high nutritional value and versatile functional properties in food products. Estimates of the worldwide production of whey indicate that about 700,000 tonnes of true whey proteins are available as valuable food ingredients. Nutritional and functional characteristics of whey proteins are related to the structure and biological functions of these proteins. During recent decades, interest has grown in the nutritional efficacy of whey proteins in infant formula and in dietetic and health foods, using either native or predigested proteins.

This paper focuses attention on the differences and similarities in composition of human and bovine milks with reference to infant formula. More desirable milk protein composition for consumption by humans is obtained by the addition of lactoferrin and more specific fractionations of proteins from bovine milk. Optimization of heating processes is important to minimize the destruction of milk components during fractionation and preservation processes.

Some functional characteristics of whey proteins are discussed in relation to their properties for application in food products. Information obtained from functional characterization tests in model systems is more suitable to explain retroactively protein behavior in complex food systems than to predict functionality.
(**Key words**: whey proteins, functional properties, nutritional properties, infant formula)

Abbreviation key: **EU** = European Union, **LF** = lactoferrin, **LP** = lactoperoxidase, **PP** = proteose-peptone content, **WPC** = whey protein concentrate.

INTRODUCTION

Bovine milk is designed for nutrition, growth stimulation, and immunological protection of the young calf. Since prehistoric times, bovine milk has been consumed by human beings either as fluid milk or other dairy products. During the last century, the production of milk in western countries has far exceeded overall consumer demand for milk and dairy products, and the surplus of bovine milk has been discovered as a potential source of functional and nutritional ingredients for many traditional and new food products. Enormous amounts of high quality milk proteins are now available for both functional and nutritional utilization in food and feed products. Estimates of the worldwide production of skim milk powder and whey solids indicate amounts of over 2.3 and 7 million tonnes, respectively (57, 68).

New developments in fractionation, modification, and preservation techniques of milk protein products have improved their functional properties (23, 24, 33, 45). The functional applications of caseinates are well known as water and fat binders in meat products (31), milk products (e.g., yogurt) (67), and recombined products (35). Whey protein products are well known as replacements for egg proteins in confectionery (46) and bakery products (17), but whey proteins are also used as functional ingredients and as milk replacers in dairy products such as ice cream (66). These applications often occur in competetive markets in which functional vegetable proteins also are available.

During recent decades, interest in the utilization of bovine milk proteins in specialty food products, such as infant formula, dietetics, and health foods, has increased (32). This paper focuses on the utilization of whey proteins as nutritional and functional ingredients in food products and starts with an overview of the availability of whey proteins compared with other surplus dairy ingredients.

PRODUCTION AND UTILIZATION OF DAIRY INGREDIENTS

Estimates of the production of milk proteins in 1995 are shown in Table 1. The volumes of skim milk powder and whey powder produced are large compared with the amounts of specialized ingredients, such as casein or caseinate, whey protein concentrate (**WPC**), and lactose. In the European Union (**EU**), these data have not changed much compared with

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TABLE 1. Estimated production of dairy ingredients in 1995.

Nonfat ingredient	EU[1]	US[2]	Oceania[3]	Rest of world[1]	World-wide	Whey solids involved
	(10^3 tonnes)					
Skim milk	1184	541	332	267	2324[4]	. . .
Casein or caseinates	140	. . .	80	25	245[1]	490
Whey powder	824[5]	648	48	115	1635	1635
Lactose	280	138	30	12	460	920
WPC 35%[6]	150	108	. . .	7	265	795
WPC >35%[7]	40	. . .	15	. . .	55	440
						. . .
Total whey solids involved						4280

[1]R. Krimpenfort, 1997, personal communication, DMV International, Veghel.

[2]American Dairy Products Institute (1). Lactose Company New Zealand (37).

[3]Australian Dairy Corporation (3).

[4]ZMP Europa Dauermilch (68).

[5]Including 110,000 tonnes demineralized whey powder.

[6]Whey protein concentrate, that is 35% protein on a total solids basis. Tonnages include delactosed, desalted whey (57).

[7]Whey protein product that is >35% protein on a total solids basis. Tonnages include whey protein isolates.

1993 data, except that a 50% increase occurred for casein and caseinates (10).

The amounts of whey solids that are produced or required for the production of the other ingredients shown in Table 1 are given in the last column and are calculated using the following criteria: the production of 1 kg of caseinate yields 2 kg of whey solids, 1 kg of lactose uses about 2 kg whey solids, whey protein concentrate that is 35% protein on a total solids basis needs 3 kg, and higher whey protein concentrates [e.g., WPC that is 60 or 80% protein on a total solids basis] need 8 kg of whey solids on the average (58).

The worldwide production of liquid whey is estimated at 118 million tonnes (57), a quantity that is equivalent to about 7 million tonnes of whey solids. Compared with 4.3 million tonnes of whey solids (Table 1), about 62% of the worldwide whey production appears to be gainfully utilized. By taking the (true) whey protein content of this surplus whey solids at 10%, it appears that about 270,000 tonnes (38%) of the 700,000 tonnes of true whey proteins are still additional available as valuable food ingredients.

The human food sector consumed about 14% of the whey powder produced in the EU in 1991 (9); this proportion is estimated at nearly 40% in the US in 1995 (1). The amount of demineralized whey powder produced in the EU in 1995 was estimated at 110,000 tonnes (R. Krimpenfort, 1997, personal communication). Demineralized whey is mainly used in infant formula. Whey protein concentrate that is 35% protein on a total solids basis is mainly utilized as skim milk replacer, both in food and feed products. The higher protein WPC (and whey protein isolates) are used exclusively in food products; dairy products are the largest end use sector, followed by bakery and meat products (11). Interest is increasing in the utilization of native and hydrolyzed milk proteins for human nutrition and health-related foods. Although usage for these sectors in Europe is still estimated at about 10% of the total volume of both caseinate and WPC, an increase in utilization is expected (10, 11, 30). Research activities are now increasely focused on the use of milk proteins as nutritional ingredients in milk that has been altered to resemble human milk and in dietetic, health, and functional foods. The composition and biological functions of milk proteins form the starting point in most of these studies.

STRUCTURE AND BIOLOGICAL FUNCTIONS OF WHEY PROTEINS

Biological functions and related protein structures reflect an important source of information on the physicochemical characteristics and functional properties of whey proteins. Much is already known about the size and shape of the main milk proteins, and it is interesting to evaluate how these characteristics are related to their biological functions. Figure 1 shows the size differences between casein micelles and whey proteins on a molecular level. The casein micelles have primarily nutritional functions, and the whey proteins are designed more as carriers for ligands and trace elements and for biological func-

TABLE 2. Composition and biological functions of whey proteins in bovine milk.

Whey protein	Weight contribution	Biological function
Major	(g/L of milk)	—— For the calf ——
β-LG	3.2	(Pro)vitamin A transfer
α-LA	1.2	Lactose synthesis
BSA	0.4	Fatty acid transfer
IgG	0.8	Passive immunity
Bioactive		—— General ——
LF[1]	0.2	Bacteriostatic agents
LP	0.03	Antibacterial agent
Enzymes (>50)	0.03	Health indicators
Proteose-peptones	≥1	Opioid activity

[1]LF = lactoferrin; LP = lactoperoxidase.

tions. The large circle indicates the size of a casein micelle having a mean diameter of 100 nm or a radius of 50 nm. The radius, diad axis, or tetrad axis of the whey proteins vary between 1.8 nm for α-LA and 6 nm for IgG (5, 6, 22, 43). Figure 1 also indicates that the much smaller sizes of the whey proteins (compared with that of casein micelles) results in a much higher numerical contribution in milk than casein micelles. In fact, whey proteins have the capability to cover the casein micelles completely, but it is unlikely that this happens. Small proteins (such as α-LA) tend to be composed of a single globular domain, which appears to be marginal for stability and overall rigidity (64). The observation that these proteins contain disulfide bridges or metal clusters (e.g., calcium in α-LA), enhancing the stability of the folded state, strengthens this premise. The longer polypeptides (such as BSA) often fold into two or more domains, and this process is related to their biological functions as binders of ligands and drugs. The Y-shape of IgG responds to the need to have several flexible, connected ligand-binding sites for activities as carriers of passive immunity in newborns of all species (22). Because intact active bovine IgG has been detected in the feces of infants (55), IgG may withstand proteolytic degradation in the intestinal tract, which is useful for preventing gastrointestinal infections of newborns.

COMPOSITION AND BIOLOGICAL FUNCTIONS OF WHEY PROTEINS

The composition and main biological functions of the whey protein fraction in milk are summarized in Table 2. The globular structure of β-LG is remarkably stable against the acids and proteolytic enzymes present in the stomach (49). This stability is relevant to the biological function of β-LG as a resistant carrier of retinol (a provitamin A) from the cow to the young calf. This biological function appears to be less important for human babies, which may explain why β-LG does not occur in human milk. β-Lactoglobulin is a rich source of cysteine, an essential amino acid that appears to stimulate glutathione synthesis, an anticarcinogenic tripeptide produced by the liver for protection against intestinal tumors (42). The biological function of α-LA is to support the biosynthesis of lactose, which is an important source of energy for the newborn. At pH ≤4.0, α-LA unfolds and is susceptible to digestion by pepsin in the stomach (44).

Bovine serum albumin (BSA) is identical to blood serum albumin and is probably transported to the milk by leaky junctions of the blood vessels in the mammary gland (26). Bovine serum albumin binds insoluble free fatty acids for transportation in the blood and is probably an important source for the production of glutathione in the liver, a peptide which should also have immunoenhancing activity for individuals testing positive for the human immunodeficiency virus (HIV) (20).

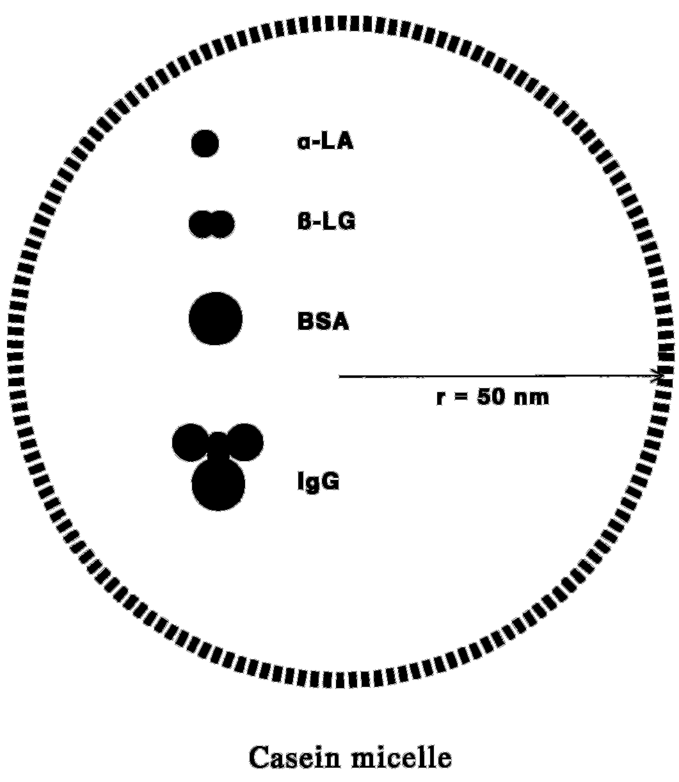

Figure 1. Size comparison of some whey proteins with a casein micelle of average size (radius 50 nm). Radius of α-LA, 1.8 nm; diad-axis of β-LG (occurring as a dimer), 3.5 nm; radius of BSA, 4 nm; and tetrad axis of IgG, 6 nm. Numbers of distinct whey protein molecules per average casein micelle in milk are α-LA, 325; β-LG (dimers), 335; BSA, 22; and IgG, 8.

The dominant species of immunoglobulins in bovine milk, IgG, is identical to blood serum IgG in virtually all mammalian species (38). Known as "pseudoglobulin" in the old nomenclature (61), IgG is the carrier of passive immunity to the newborn. The classical "euglobulin" fraction contains secretory IgA and IgM, from which IgM is related to the well-known cold creaming phenomenon of raw milk.

Well-known bioactive proteins in milk are lactoferrin (**LF**), lactoperoxidase (**LP**), and endogenous enzymes, each of which is present in minor quantities. Two main biological functions have been assigned to LF (an iron-containing protein): the antibacterial activity in the mammary gland, and the nutritional activity of making iron more available for absorption in the gut (56). However, results of in vitro and in vivo studies on both effects are not conclusive (39). Lacteroperoxidase functions biologically in the mammary gland and in the digestive tract of the calf as part of a bactericidal system (LP system) that is active against a number of enteric bacterial strains. Both LF and LP have been reported to have beneficial actions in reducing the incidence of chronic diarrhea (39, 48).

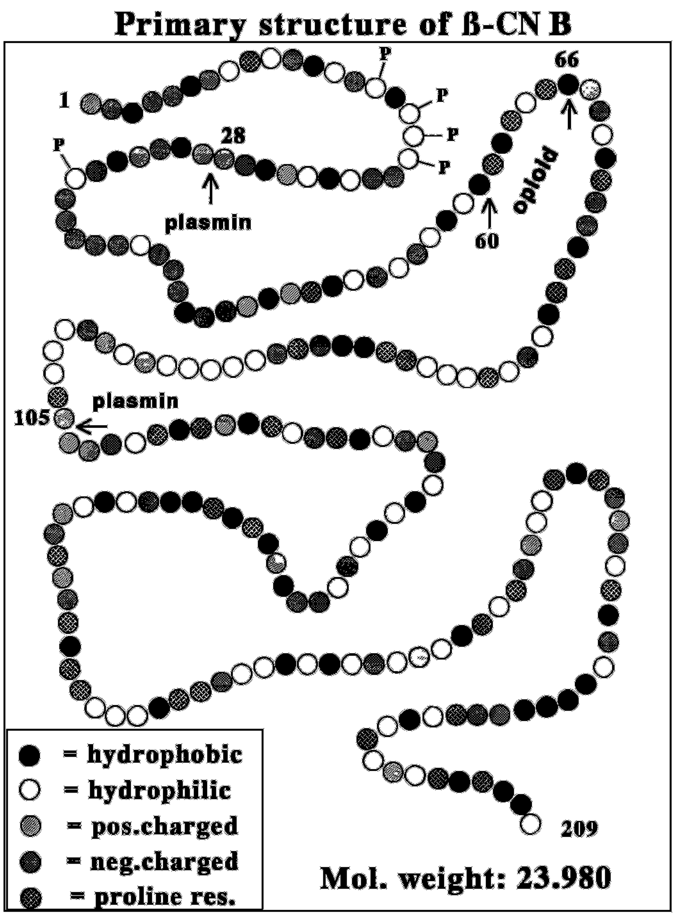

Figure 2. Primary structure of β-CN B, indicating bonds cleaved by plasmin for release of proteose-peptone (PP): PP 8–fast (residue f1–28), PP 8–slow (f29–105/107), and PP 5 (f1–105/107). pos. = Positively, neg. = negatively, res = residue, mol. = molecular, and P = phosphoseryl residue.

Over 50 different enzymes have been reported to be indigenous to milk, and most enzymes originate from mammary tissue cells, blood plasma, and leucocytes, which indicates that the biological functions of the milk enzymes are quite different, although their occurrence and properties appear to be similar in all mammals (2).

The proteose-peptone fraction, often regarded as part of the whey protein fraction, is composed mainly of polypeptides from β-CN through actions of proteinases, particularly plasmin. The origin and storage time of fluid milk may be responsible for a highly variable proteose-peptone (**PP**) content, ranging from 1 to 3 g/L (2). Figure 2 shows the primary structure of β-CN B, indicating the locations (f28 and 105 or 107) of plasmin attacks. The resulting polypeptides up to amino acid 105 are known as proteose peptone fractions PP-5 (amino acid f1–105), PP-8 fast (f1–28), and PP-8 slow (f29–105). Fraction PP-8 fast is a phosphopeptide, which may enhance the gastrointestinal absorption of calcium (36). Fraction PP-8 slow contains peptides with opioid activity, such as β-CN (amino acids f60–66), as indicated in Figure 2, known as β-casomorphins, which have a number of physiological effects (59).

ADAPTATION OF BOVINE MILK FOR INFANT FORMULA

During the past few decades, interest has grown in improving the nutritional and biological performances of milk that has been modified to resemble human milk for use in products such as infant formula. For that purpose, ingredients from bovine milk are isolated or adapted using human milk as reference. The composition of bovine milk, although it differs from human milk in many respects, is still the most selected nutrient source for infant formula.

Comparison of the composition of human and bovine milks reveals that bovine milk has much higher casein and mineral contents than does human milk. One of the first adaptations was, therefore, to supplement bovine milk with desalted whey or WPC to produce infant formulas that were based predominantly on whey proteins. The ratio between caseins and whey proteins is then reduced from 80:20 in bovine milk to 40:60 in infant formula. In these formulas, caseins may be present either as casein

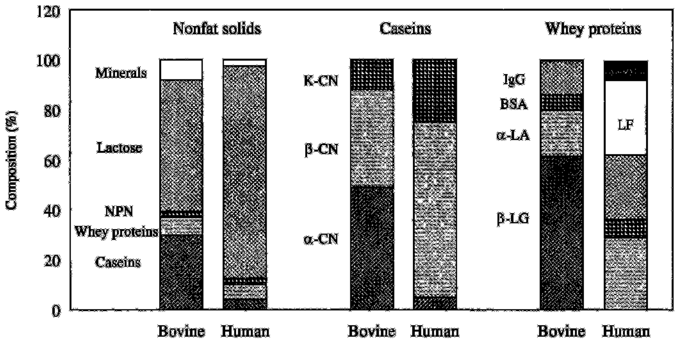

Figure 3. Compositional differences of nonfat milk solids, caseins, and whey proteins between bovine and human milks (LF = lactoferrin).

micelles or as caseinate, depending on the requirements for calcium and phosphorus. In bovine milk, about 44% of the inorganic phosphate is associated with casein micelles as calcium phosphate (29); in human milk, only 23% is associated as calcium phosphate (27). Excessive phosphorus intake may contribute to hypocalcemia in newborn infants (18). In situations in which hypocalcemia is a possibility, bovine milk in infant formula may be replaced by formula based on calcium caseinate and whey proteins. Manipulation of the ratio of casein to whey dilutes the remaining milk solids so that specific essential nutrients have to be added. The NPN content has to be adapted, especially when WPC has been used as the whey protein source (16). Lactose has to be added, and the fatty acid composition has to be altered, to yield a higher unsaturated fat content (50). Also, the mineral content needs adaptation to the required osmolar load (51). Some trace elements (e.g., iron, copper, and manganese) are supplemented (34), and vitamins have to be added (21).

One may wonder whether the differences between human and bovine milks are really so large as these adaptations suggest. In fact, the main cause for the distinctions between these milks are related to differences in growth rates and transfer routes for maternal immunity to the newborns (calf and infant).

THE COMPOSITION OF HUMAN AND BOVINE MILKS

A striking difference between bovine milk and human milk is that, although the total solids contents are similar, bovine milk has about a 7 times higher casein content and a 3 times greater mineral content than human milk (Figure 3, columns 1 and 2). Removal of the surplus amounts of caseins and minerals from bovine milk increases the lactose content from 50 to over 75% on fat-free solids, which approaches that of human milk. Removal of surplus caseins is, in fact, another approach for changing the ratio between caseins and whey proteins from 80:20 to 40:60.

Comparison of the casein composition of both milks (columns 3 and 4) indicates that a quantitative removal of α_S-CN increases the proportion of β-CN from 35% to nearly 75% (the content of human milk). The physiological impact of α_S-CN in infant formula is the production of a clot in the stomach that is too firm (47). Attempts to remove α_S-CN from the casein fraction have already been described (63). Two-third of the salts in bovine milk are in the colloidal phase and are related to casein micelles. Removal of most of the casein micelles also reduces the salt content of bovine milk significantly. The relative contributions of sodium, potassium, calcium, and magnesium to the mineral content differs very little between bovine and human milks.

Removal from bovine milk of β-LG, which is not present in human milk, increases the α-LA content from 18 to 40% of the whey proteins (see columns 5 and 6), which is identical to that of human milk. So bovine milk contains a surplus rather than a shortage of nutritional components compared with human milk. Some important deficiencies are the presence of LF (about 25% of the whey proteins) and lysozyme (5% of the whey proteins) as major proteins in human milk and the Ig predominantly present as IgA instead of IgG. Most of the enzymes, growth factors, and vitamins occur at nearly identical levels in both milks and need not be removed.

Currently, research activities are focused on the fine-tuning of both milk protein and amino acid composition. By enriching bovine milk with α-LA, β-CN, and LF, the optimal nutritional composition may be achieved, particularly in amino acid composition. Another approach might be the separation of surplus components from bovine milk by using physical separation techniques. Some reports (8, 60) have discussed the solubilization of β-CN in cooled milk, and a number of patents have subsequently claimed to separate either casein micelles from milk or α_S-CN from caseinates (62, 63, 65).

A theoretical preparation based upon mixing β-LG depleted whey proteins, β-CN, and LF in desalted (milk) permeate reveals that the protein composition more closely approaches that of human milk than that of existing infant formulas. Figure 4 shows the composition of this preparation compared with bovine milk, human milk, and an existing infant formula. In the theoretical formula, K-CN is not present but is

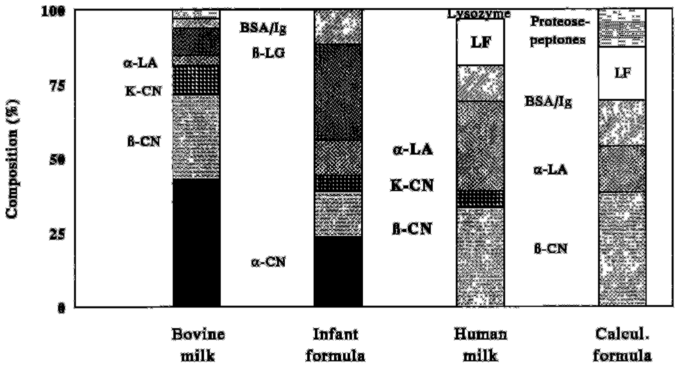

Figure 4. Protein composition of bovine milk, infant formula containing 60% whey proteins and 40% caseins, human milk, and a theoretical infant formula.

replaced by additional β-CN. The proteose-peptone fraction, which is normally present in bovine milk, is also introduced in the theoretical formula. As discussed already, proteose-peptone contains both calcium phosphopeptides and opioid peptides, which may have additional nutritional effects. The result is nearly 90% homology in protein composition with human milk; existing formula have only about 70% homology with human milk.

AMINO ACID COMPOSITION OF INFANT FORMULA

Much attention has been paid to the nutritional aspects of the essential amino acid composition of formulas and the use of protein hydrolysates. Protein hydrolysates are generally used in enterally administered nutritionally complete formulas that are given to individuals who have specific nutritional or physiological needs or are used as nutritional supplements for conventional foods. Protein hydrolysate composition is generally divided into three major groups: 1) formulas for infants with allergies to intact proteins, 2) elemental diets for patients with impaired gastrointestinal functions, and 3) nutritional supplements to provide nitrogen in an easily assimilated form (41). Whey protein hydrolysates have been used as primary nitrogen source in infant formula with reduced allergenic properties.

The amino acid composition of formulas appears to be a critical factor for the nutrition of newborn infants when intact proteins or protein hydrolysates are used. Figure 5 shows a comparison of the 10 essential amino acids in an existing infant formula (60% whey proteins, 40% caseins) and human milk (28). The most important differences are the lower concentrations of cysteine and tryptophan, which are considered to be vitally important for physiological pathways, as, for example, the prevention of oxidative damage by cysteine as a component of glutathione, and tryptophan as precursor for the neurotransmitter serotonin, which induces sleep (53). The lower concentrations of tryptophan and cysteine appear to be a major problem for the production of suitably balanced amino acid profiles in blood plasma in infant formulas. Total whey proteins do have not the ability to correct the balance of essential amino acids completely (54). Only bovine α-LA having much higher cysteine and tryptophan contents than human milk proteins is recommended as supplementary (whey) protein in infant formula (28).

The concentrations of some other essential amino acids, such as lysine and threonine, are significantly higher in infant formula than in human milk. The increased threonine content in formulas may even double the concentration of plasma threonine compared with that of human milk (57), which hampers the possibilities of getting optimal amino acid profiles in blood plasma, being the golden standard for newborn infants (52). The increased plasma threonine content has been assumed to be caused mainly by the presence of glycomacropeptide from cheese whey (19).

The amino acid composition of the theoretical formula based upon selected components of bovine milk (Figure 4) reveals a much more balanced composition (Figure 6). The most important essential amino acids, cysteine, methionine, and tryptophan, are present in slightly higher amounts, and the surplus of threonine is reduced to 10%. The contributions of isoleucine and tyrosine are still too low because the concentration of β-CN in the theoretical formula is too high, but this situation may be easily corrected.

In addition to the efforts for designing a representative composition of a milk that is closer to human

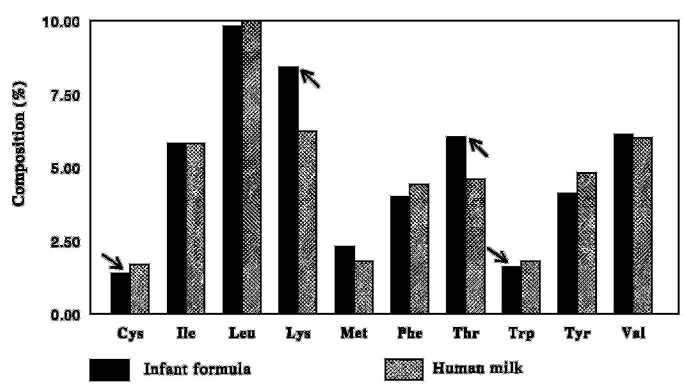

Figure 5. Differences in essential amino acids (as a percentage of total amino acids) between human milk and infant formula.

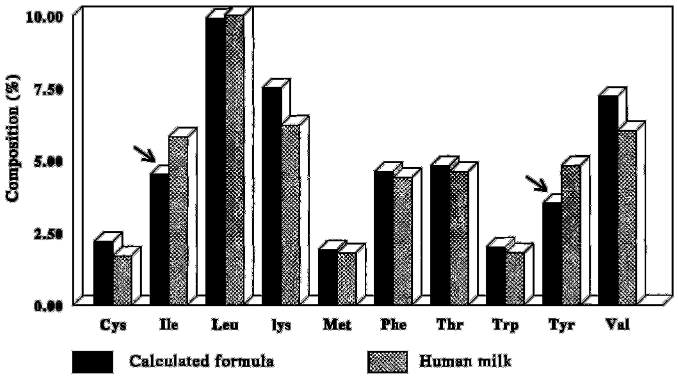

Figure 6. Differences in essential amino acid composition (as a percentage of total amino acids) between human milk and the theoretical formula.

milk, the destruction of milk components during processing and preservation should be minimized. A number of new techniques for preservations under mild conditions have appeared, such as high pressure pasteurization (40), infuse sterilization (12), and microfiltration (23). Most of them are in the developmental stage and are not generally applied. Optimization of the classical heating processes is still important to consider, particularly for whey proteins.

EFFECTS OF HEAT TREATMENT ON BEHAVIOR OF WHEY PROTEINS

The denaturation of whey proteins may be considered as a two-step process: 1) unfolding, which may be reversible or irreversible, and 2) aggregation, which generally follows irreversible unfolding (14). To be able to optimize heat treatments that ensure the proper functional and nutritional uses of whey protein products, detailed information is needed about the kinetics of unfolding and aggregation of whey proteins.

Unfolding of globular proteins is accompanied by an endothermal heat effect, which may be observed by differential scanning calorimetry. The shape of the heat absorption curve gives information on the kinetics of the unfolding process and can be analyzed according to the well-known method of Borchard and Daniels (4). Protein unfolding appears to be a first-order reaction.

The rate of protein aggregation may be observed by using high performance gel permeation chromatography in buffer at pH 6.7. This technique allows the measurement of residual concentrations of native proteins both before and after the selected heat treatment. Differences in these concentrations, measured as a function of time at various selected temperatures, reveals the kinetics of aggregation (14). Figure 7 shows calculated results of unfolding and aggregation kinetics obtained for β-LG during a continuous heating process of whey at two extreme heating rates: a heating rate of 1°C/s, which may occur in the heating section of industrial pasteurizers, and a heating rate of 1°C/min, which takes place in the regenerative sections of heat exchangers with extensive facilities for heat recovery. The lines in Figure 7 indicate the net availability of unfolded β-LG, and the bars indicate the amount of aggregated protein over the indicated temperature range. At both heating rates, aggregation (i.e., irreversible denaturation) appears to start when 60% unfolding has occurred. This result implies that a heating rate of 1°C/s does not show aggregation below 83°C, but, at a heating rate of 1°C/min, aggregation has started already at 73°C.

Heat exchangers using different heating rates may have a significant effect on the extent of whey protein denaturation during heating to, or cooling from, the selected holding temperature. To control thermal denaturation in heat exchangers, the holding time at a selected temperature (a characteristic parameter) may be better replaced by an equivalent heating time at a selected (holding) temperature, usually indicated as the *F* value. The extent of differences in heat load in industrial pasteurizers having heat recoveries of 80 or 95% is shown in Figure 8. The holding time was adjusted at 10 s for both pasteurizers at the indicated holding temperatures. The arrows indicate the *F* values, when 75, 80, 85, or 90°C were used as holding temperatures. Clearly, the pasteurizer using 80% heat recovery operated with an effect identical to a holding time of more than 12 s at the indicated

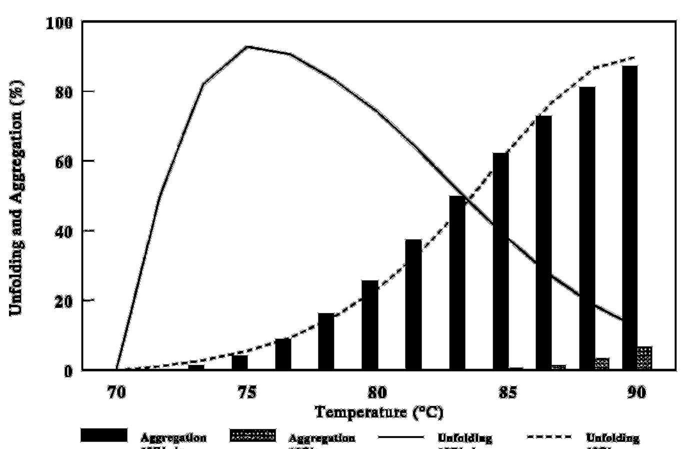

Figure 7. Percentage of unfolding and of aggregation of β-LG in whey during heating processes between 70 and 90°C at various heating rates.

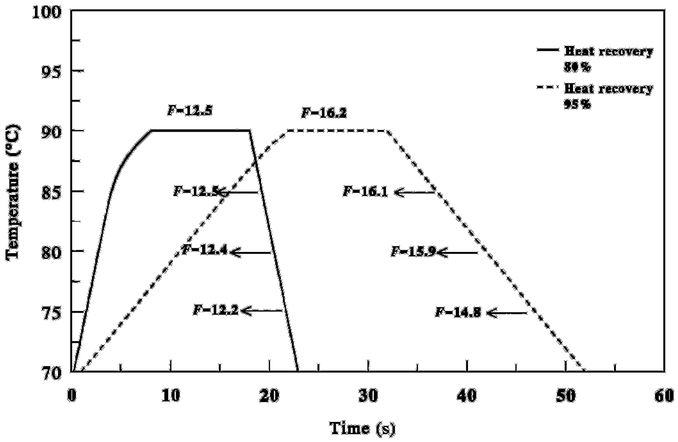

Figure 8. Temperature and time curves for pasteurizers (25,000 L/h) with 10 s holding time, using heat recoveries as indicated. The F values indicate the calculated holding times at the specified temperatures.

holding temperatures; the pasteurizer using 95% heat recovery shows a F value of more than 16 s at 85°C.

The effects of these different heating processes on the denaturation of β-LG in whey and a native whey protein isolate (BIPRO; Le Sueur Isolates, Le Sueur, MN) are shown in Figure 9. Significant differences in protein aggregation (using heat recoveries of 80 and 95%) appear at 10 s of holding time above 75°C. In BIPRO, β-LG aggregation increases to 7 and 10% on heating processes adjusted to 10 s at 90°C in these pasteurizers, respectively. The same heat treatments, performed at identical concentrations of β-LG in whey, result in a denaturation at 90°C of as much as 23 and 28%, respectively. This feature is caused by the dominant effects of salts, particularly calcium ions, on the denaturation of whey proteins. The heat treatments shown in Figure 9 are representative for most heating processes used during evaporation and drying of whey protein products. The irreversible denaturation effects of β-LG are of the same magnitude as those observed for most whey protein products and are directly reflected in their functional properties.

EFFECTS OF PROTEIN DENATURATION ON FUNCTIONAL PROPERTIES OF WHEY PROTEIN PRODUCTS

Functional properties as determined by arbitrarily selected techniques (13) for β-LG and seven industrially prepared whey protein products are shown in Figure 10. The functional properties are presented on a relative basis using a scale of 0 and 100%, so that the results obtained would be easier to discuss; 100% refers to the best functional characteristics ever obtained by that particular technique using a milk protein product (caseinate and total milk protein concentrate included). Experimental results of four functional properties are presented in a cumulative way; that is, 400% should reflect the most functional protein product. The products indicated as 2, 3, and 4 were obtained by ionic exchange processes, and 5, 6, and 8 were obtained by membrane processes.

The solubility at pH 4.6 appears to be 100% only for β-LG, and most of the industrially prepared samples show solubility characteristics between 80 and 90%. Reduced solubility (determined at pH 4.6) reflects whey protein denaturation, which usually is caused by the evaporation and drying steps of whey protein products. Products 4 and 8 exhibit more serious denaturation because they have been evaporated and dried under different environmental conditions, namely, at pH 8 and 3.5, respectively. Figure 10 clearly shows that the solubility at pH 4.6 is closely related to the other functional properties, and this result stresses the need for controlled heat treatments during processing and drying to keep the whey proteins in their native state.

RELATIONSHIP BETWEEN FUNCTIONAL PROPERTIES AND PROTEIN FUNCTIONALITY IN FOOD PRODUCTS

The term functional properties is often used in relation to physicochemical properties of proteins in aqueous solutions or in simple model systems. The functional behavior of milk proteins in food products,

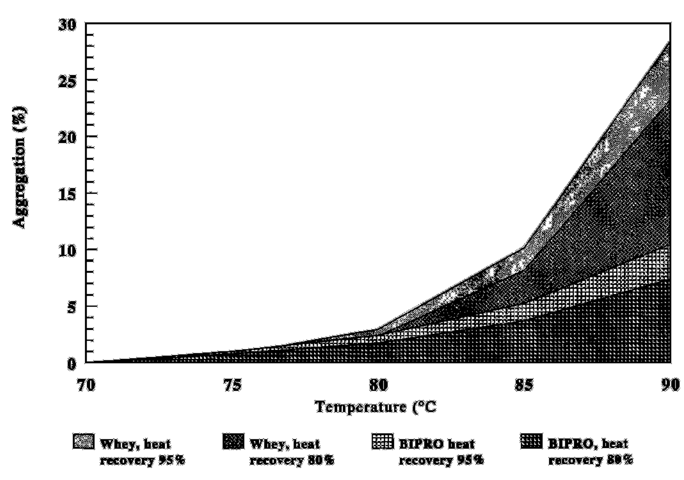

Figure 9. Effect of heating for 10 s at the indicated temperatures on the aggregation of β-LG in whey and native whey protein isolate (BIPRO; Le Sueur Isolates, Le Sueur, MN) , using pasteurizers with the indicated heat recoveries.

however, is much more complicated (Figure 11). The native proteins, as synthesized by the cow (top portion of Figure 11), reflect a number of functional properties in aqueous solutions (middle portion, Figure 11), which are modified during food processing to yield the protein functionality (bottom portion, Figure 11). In this sequence, the functional properties are the result of intrinsic properties of native proteins and a number of extrinsic factors. Intrinsic factors include amino acid composition, amino acid sequence, conformation, molecular size, flexibility, net charge, and hydrophobicity of protein molecules. Knowledge of the relationship between intrinsic protein properties and extrinsic factors such as temperature, pH, salts, and protein concentration is critically important for elucidating and controlling the functional properties. These functional characteristics, however, are informative primarily about the process history and composition of the whey protein product involved. The performance of whey proteins in food products is another characteristic that may be defined as protein functionality. This characteristic reflects the manner in which proteins interact with components of a food product, such as lipids, sugars, salts, and other proteins, as indicated in Figure 11. These interactions are governed by the effects of processing, such as homogenization, heating, freezing, and storage conditions. Thus, almost all applications require specific functional attributes to obtain the desired performance, which is usually achieved by trial and error. To improve the predictability of functional properties, solid relationships between the various factors and effects shown on Figure 11 should be obtained.

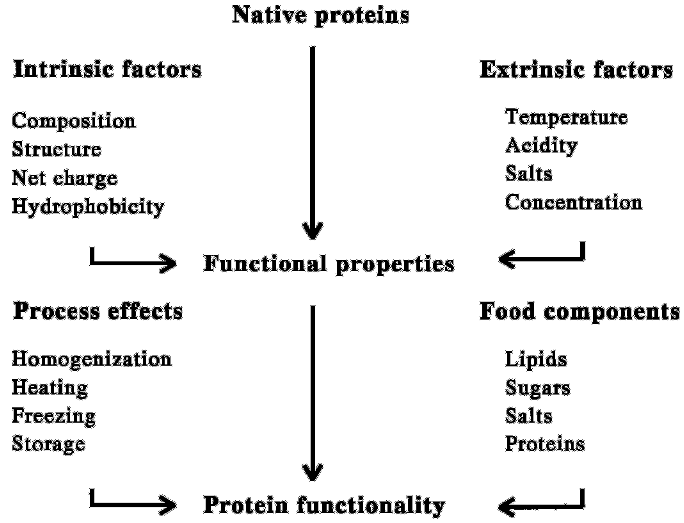

Figure 11. Driving forces involved in achieving functional properties and protein functionality of whey proteins.

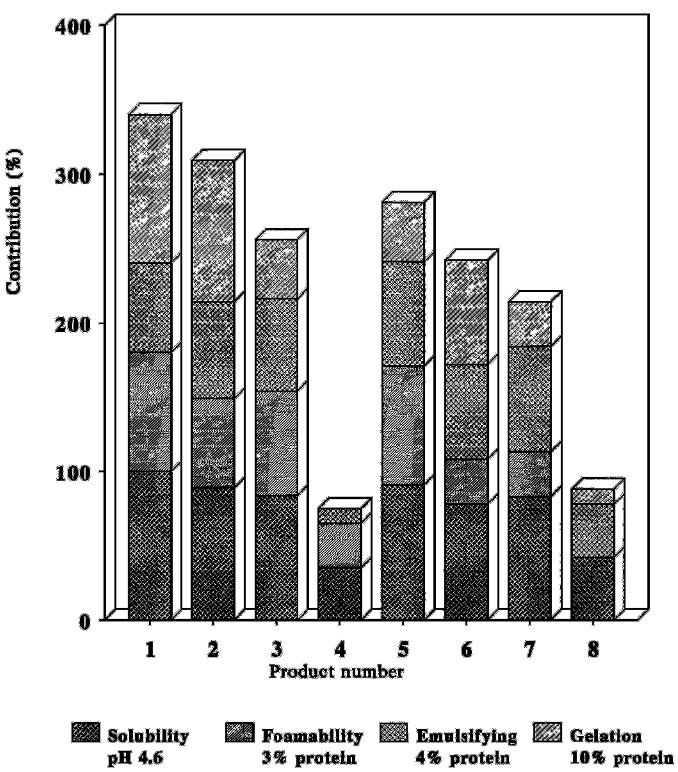

Figure 10. Comparison of some functional properties of β-LG with those of industrially prepared whey protein products: 1) β-LG, 2) fixed bed ion-exchange whey protein isolate (WPI) (Spherosil QMA) Fromageries BEL, Vendôme, Fr. 3) stirred bed ion-exchange WPI (BIPRO; Le Sueur Isolates, Le Sueur, MN), 4) fixed bed ion-exchange WPI (Spherosil S), 5) defatted UF WPC, 6) neutral UF and diafiltration whey protein concentrate (WPC), 7) normal UF WPC, and 8) acid UF and diafiltation WPC.

PROTEIN FUNCTIONALITY IN FOOD PRODUCTS

It is very difficult to find solid relationships between functional properties and behavior of proteins in a food product, which can be illustrated by some examples. An important functional demand of whey proteins is their suitability to induce and stabilize aerated food products. This property is usually identified with the foaming ability of whey protein products in aqueous solution. It is generally known that the presence of a small amount of fat in a WPC (even after centrifugation of the whey) is detrimental to its foaming properties, and this effect is also clearly shown by meringues. Meringue is normally prepared from egg white to which sugar is gradually added during the whipping process. This whipped sugar preparation is dried at 100 to 110°C and puts high

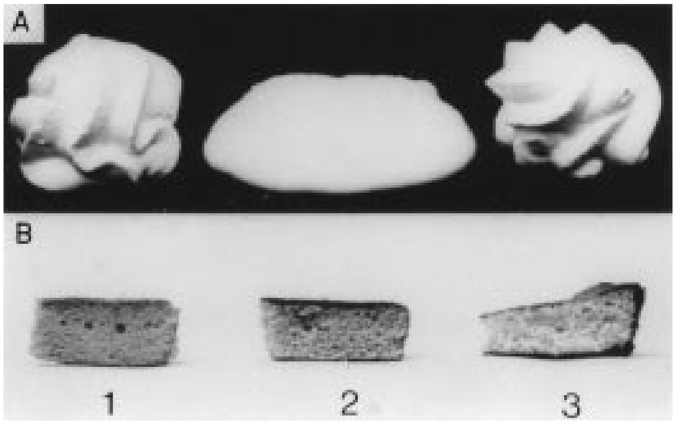

Figure 12. A. Meringues prepared from: (left to right) sugar and egg white, normal UF whey protein concentrate (WPC) 60% protein, and 3) defatted UF WPC 60% protein. B. Sponge cakes were prepared from 1) sugar, flour, and whole egg; 2) normal UF WPC; and 3) defatted WPC.

demands on foam stability during the drying process. Meringue prepared from defatted UF WPC (<1% fat on protein) is of a similar structure and appearance as the reference egg white product (Figure 12A). Meringue prepared from a normal UF WPC (5% fat on protein) is reduced to a flat biscuit during the drying process. An opposite result, however, is observed in whipped products containing sugar and wheat flour such as sponge-type cakes when whole egg is replaced by WPC (Figure 12B). The WPC that had not been defatted apparently produced results that were superior to those of the defatted WPC, which may be explained by a different type of foam obtained in whey that had not been defatted. This phospholipid and protein foam (7) appeared to be more compatible with the wheat flour used in sponge-type cakes.

The discrepancy between foaming properties and aeration properties of whey proteins is even greater in Madeira-type cakes, consisting of protein, sugar, flour, and butter. The emulsifying or fat-binding properties of whey proteins are now much more important than their foaming performances. Figure 13 shows cakes that were prepared using whole eggs and UF WPC as complete substitute for whole eggs. The specific volume (aeration) of the batter of UF WPC cake prepared according to the normal procedure was strikingly similar to that of the whole egg cake. However, during the second baking stage, the UF WPC cake collapsed completely under fat separation, indicating that fat impairs the structure-forming properties of whey proteins during baking. Previous emulsification of the fat in UF WPC resulted in cake

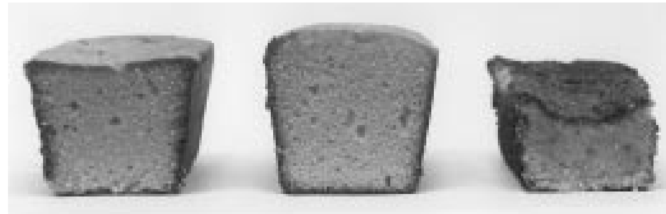

Figure 13. Madeira-type cakes, prepared from (left to right) equal amounts of wheat, flour, sugar, fat, and whole eggs; UF whey protein concentrate (WPC) in preemulsion, and UF WPC in normal recipe.

that had an even higher specific volume than the reference whole egg cake after baking (Figure 13). Thermally induced structure-forming properties are, in fact, the most important functional properties of whey proteins and may form the basis for quite new product structures, as is illustrated with the following product examples.

WHEY PROTEINS AS ESSENTIAL FOOD INGREDIENTS

The good emulsifying properties of whey proteins allow the introduction of fat globules as structural elements of heat-induced whey protein gels. Moreover, the well-known heat-induced interactions between whey proteins and casein micelles make milk an interesting base for all kinds of textured products with high nutritional value.

Figure 14 shows examples of new food product structures based on thermally induced interactions of milk proteins (15). From left to the right, product structures are shown that simulate cheese, meat, and confectionery products obtained after heat treatment of model systems. The sliceable cheese structure (Figure 14) is obtained by heating an emulsion of milk fat in UF WPC. During the heating process, air

Figure 14. New food structures based on thermally induced interactions of whey proteins. Cheese-like, meat-like, and confectionery-like product structures were obtained by heating an emulsion of milk fat in UF whey protein concentrate (WPC), in defatted WPC or in skim milk enriched with whey proteins (see text).

is injected to obtain some variation in the thus obtained sliceable structure. The second cheese-like product (Figure 14) is somewhat softer and easier to cut. This structure is obtained by heating an emulsion containing skim milk solids, milk fat, and whey proteins. To improve its eating quality, this structure was interrupted by zones of walnuts to obtain some similarities with a French-type Rambol cheese. The composition of meat-like structures supplemented with 1.5% salts and meat flavor components (Figure 14) is nearly identical to that of the cheese-like structures. The eating quality of these structures is improved by incorporation of either small fibers or pieces of smoked sausage. Confectionery-like products (Figure 14) are prepared from (defatted) UF WPC, shown by the meringue, or from UF skim milk, shown by the high protein bar with chocolate coating. These protein structures are not tailor-made products, but they illustrate the technological possibilities that may be achieved using undenatured whey proteins combined with heat treatments during product manufacture.

CONCLUSIONS

A source of about 700,000 tonnes of whey proteins is available for use as valuable food ingredients, of which about 62% appears to be gainfully utilized. The biological functions and related structures of whey proteins reflect the important functional characteristics of the milk protein products. Compared with human milk, bovine milk contains a surplus rather than a shortage of the major nutritional ingredients. Infant formula may be adapted to compositions more resembling human milk by removing surplus ingredients from bovine milk and adding ingredients such as LF, some vitamins, trace elements, and adapting the fat composition.

Functional properties of whey proteins are governed by the composition and structure of the protein and influenced by the prevailing environmental conditions, prior treatments, processing conditions, and methods for their characterization, which makes it extremely difficult to evaluate these properties and to predict their effects on the functionality of proteins in food products. Information obtained from functional characterization tests in model systems is more suitable to explain protein behavior retroactively in complex food systems than to predict their functionality.

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